MATH 2411 Applied Statistics, L1, Spring 2024

1. Course Description

Credit Points: 4
Pre-requisite: A passing grade in AL Pure Mathematics / AL Applied Mathematics OR MATH 1014 OR MATH 1020 OR MATH 1024
Exclusion: IELM 2510, ISOM 2500, LIFS 3150

Brief Information:
This course covers the basics of probability theory, including random variables, probability distributions, and expectation, and provides an introduction to statistical inference, including point and interval estimation, hypothesis testing, ANOVA, linear regression, and using R to perform these analyses.

2. Instructor and Lectures

L1 Lectures

Name: Prof. Yu HU

Office: Rm 3443 Email: mahy@ust.hk (Please include [MATH 2411] in title for all emails.)

Date/Time: Tue & Thu 4:30 PM – 5:50 PM **Venue**: LTE

Office Hour L1:

Date/Time: TBA (will post on Canvas) **Venue:** Rm 3443, Lift 25-26

Or email the instructor to make an appointment.

3. Teaching Assistant and Tutorials

T1A (the first session is on Feb 5)

Name: HOU, Zhen

Email: zhouah@connect.ust.hk

Date/Time: Wed 04:30PM - 05:20PM **Venue:** Multi-function Room, LG4, LIB

T1B (the first session is on Feb 10)

Name: HUANG, Xinrui

Email: xhuangcn@connect.ust.hk

Date/Time: Mon 04:00PM - 04:50PM **Venue**: Rm 2504, Lift 25-26

4. Intended Learning Outcomes

Upon successful completion of this course, students should be able to:

No.	ILOs
1	Solve basic problems in probability.
2	Make inferences about the population by applying suitable statistical approaches, such as estimation, hypothesis testing, and linear regression.
3	Formulate statistical questions for real-world problems and interpret the results.
4	Analyze data using R.

5. Assessment and Grading

This course will be assessed using criterion-referencing and grades will not be assigned using a curve. Detailed rubrics for each assignment are provided below, outlining the criteria used for evaluation.

6. Grading Schemes

a. Examination duration: 2 hrs for the midterm exam and 2 hrs for the final exam
b. The **maximum score** from the two grading schemes below will be used to determine the student's final grade of the course.

Scheme 1:

Assessment	Assessing Course ILOs
30% by homework (five sets)	1, 2, 3, 4
30% by the midterm exam	1, 2, 3, 4
40% by the final exam	1, 2, 3, 4

Scheme 2:

Assessment	Assessing Course ILOs
30% by homework (five sets)	1, 2, 3, 4
0% by the midterm exam	1, 2, 3, 4
70% by the final exam	1, 2, 3, 4

The midterm exam will cover the topics in about the first part of the course until and including Estimation. The final exam will cover ALL topics of the course, with an emphasis on the second part taught after the midterm exam. All exams are closed book, with basic calculators and one A4 paper formula sheet allowed.

c. Late submissions of homework beyond 24 hours past due will not be accepted (0 score); late submissions within 24 hours will result in a percentage penalty in score.
d. No make-up midterm exams will be arranged for any reason. Student missing the midterm exam will be graded by Scheme 2 only.
e. Make-up final exams will only be arranged for exceptional circumstances, such as medical emergence with **a medical certificate issued on the same day** by a registered medical practitioner. For cases due to other particular reasons, appropriate supporting documentation is also required. ALL documents/reasons provided are subject to the verification of the University.
The student needs to **fill an application form according to University's policy within one week**: https://registry.hkust.edu.hk/resource-library/examination-regulations-student

Extenuating Circumstances Affecting Assessment

Illness or Other Circumstances Affecting Assessment
(Academic Regulations : UG Regulation:7.5; PG Regulation: 7.4)

If students wish the University to take into account illness or some other extenuating circumstances that have affected their performance in an examination, or ability to attend an examination, or to complete other assessment activities, they must report the circumstances of the case in writing and provide appropriate documentation to the Academic Registry within one week of the scheduled date of the assessment activity. The Academic Registrar will review the case and make a recommendation to the relevant Dean or the Dean's designee.

Please click the link below to download the standard report form.

[Form EX-16] Report on Extenuating Circumstances Affecting Assessment.pdf

Final Grade Descriptors:

Grades	Short Description	Elaboration on subject grading description
A	Excellent Performance	Demonstrates mastery of all probability theory and statistical analyses covered in the course and can adapt and apply the statistical framework to new problems and situations. The student can appropriately apply these concepts to real-world problems, interpret the results accurately, and exhibits strong proficiency in using R for data analysis.
B	Good Performance	Shows a solid understanding of the probability theory and statistical analyses discussed in the course. The student can effectively apply these methods to real-world problems and interpret the results with minor errors. Demonstrates good competency in using R for performing analyses.
C	Satisfactory Performance	Possesses adequate knowledge of the core probability and statistical concepts covered in the course. The student can address familiar problems and perform basic analyses using R, though with some inconsistencies. Shows basic ability to interpret results.
D	Marginal Pass	Has a threshold understanding of the fundamental probability and statistical concepts taught in the course. The student can perform simple analyses using R but may struggle with more complex applications. Interpretation of results may be limited and occasionally inaccurate.
F	Fail	Does not have sufficient understanding of even the basics components of the probability theory and statistical analyses covered in the course. The student is unable to effectively apply methods to real-world problems, nor using R for analyses, and fails to interpret results correctly.

7. Course AI policy

Students are permitted to consult any person—including the instructor, teaching assistants (TAs), and classmates—for ideas and hints while completing homework assignments. The use of ChatGPT and other generative AI tools is also allowed. However, students are required to write up the solutions independently and are responsible for ensuring that their submissions are correct and comply with University rules and laws, including those regarding plagiarism.

Students are particularly cautioned about the potential inaccuracies and fallacies that may arise from AI-generated answers. Additionally, all examinations are closed book, meaning that AI tools and resources will not be available during exams. This policy is in place to prevent overreliance on such tools and to ensure that assessments accurately reflect each student's individual understanding and capabilities.

8. Communication and Feedback

Assessment marks for individual assessed tasks will be communicated via Canvas within two weeks after the due date. Feedback on assignments will include scores and solutions. Students who have further questions about the feedback including marks should consult the instructor within five working days after the feedback is received.

9. Student Learning Resources

No textbook is required and the main reference is the lecture slides, which will be made available on Canvas progressively before each lecture.

Optional textbook for additional examples:
"*Probability and statistical inference*" by Hogg et al. Available in HKUST library.

10. Course Schedule

- Introduction
- Basic probability theory
- Random variables, distribution, special random variables
- Point estimation, confidence interval
- Hypothesis testing, p-value, t-test
- Two sample tests, ANOVA
- Linear regression

11. Academic Integrity

Students are expected to adhere to the university's academic integrity policy. Students are expected to uphold HKUST's Academic Honor Code and to maintain the highest standards of academic integrity. The University has zero tolerance of academic misconduct. Please refer to Academic Integrity | HKUST – Academic Registry for the University's definition of plagiarism and ways to avoid cheating and plagiarism.